

Technical Report

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1 Preliminaries

Graph. Let $G = (V, E)$ be a graph, where $V = \{v_1, v_2, \dots\}$ is the set of nodes and $E = \{e_{ij} \mid v_i, v_j \in V\}$ is the set of edges connecting these nodes. We use $n = |V|$ to denote the number of nodes in G , and $m = |E|$ to denote the number of edges in G .

Laplacian Energy. The Laplacian matrix $\mathbf{L}$ of a graph G is defined as the difference between its degree matrix and its adjacency matrix. The Laplacian energy $\mathcal{L}(G)$ of G is defined as the sum of the squares of the eigenvalues of $\mathbf{L}$:

$$\mathcal{L}(G) := \sum_{i=1}^n \lambda_i^2, \quad (1)$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of $\mathbf{L}$. The Laplacian energy indicates the connectivity variation of the graph.

2 Laplacian Energy Decomposition

Lemma 1 (Laplacian Energy Decomposition). *Let $G = (V, E)$ be a graph with n nodes and m edges. The Laplacian energy $\mathcal{L}(G)$ of G satisfies the following property, where $\deg(v)$ denotes the degree of v and w_e is the weight of edge e :*

$$\mathcal{L}(G) = \sum_{v \in V} \deg(v)^2 + 2 \sum_{e \in E} (w_e)^2. \quad (2)$$

Proof. By definition, the Laplacian energy of a graph G is the sum of the squares of the eigenvalues $\lambda_1, \dots, \lambda_n$ of its Laplacian matrix $\mathbf{L}$. That is:

$$\mathcal{L}(G) := \sum_{i=1}^n \lambda_i^2. \quad (3)$$

Since the Laplacian matrix $\mathbf{L}$ is real and symmetric, it is guaranteed to be orthogonally diagonalizable. Specifically, there exists an orthogonal matrix $\mathbf{Q}$ (satisfying $\mathbf{Q}\mathbf{Q}^T = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}$)

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and a diagonal matrix $\mathbf{D}$ such that $\mathbf{L} = \mathbf{Q}\mathbf{D}\mathbf{Q}^T$. The diagonal entries of $\mathbf{D}$ are precisely the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\mathbf{L}$.

Using this property, we can express $\mathbf{L}^2$ as:

$$\mathbf{L}^2 = (\mathbf{Q}\mathbf{D}\mathbf{Q}^T)(\mathbf{Q}\mathbf{D}\mathbf{Q}^T) = \mathbf{Q}\mathbf{D}^2\mathbf{Q}^T. \quad (4)$$

Next, we consider the trace of the matrix $\mathbf{L}^2$. Leveraging the cyclic property of the trace ($\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB})$), we have:

$$\text{tr}(\mathbf{L}^2) = \text{tr}(\mathbf{Q}\mathbf{D}^2\mathbf{Q}^T) = \text{tr}(\mathbf{Q}^T\mathbf{Q}\mathbf{D}^2) = \text{tr}(\mathbf{D}^2). \quad (5)$$

Since $\mathbf{D}^2$ is a diagonal matrix with diagonal entries $\lambda_1^2, \dots, \lambda_n^2$, its trace is the sum of these entries. Thus,

$$\text{tr}(\mathbf{L}^2) = \sum_{i=1}^n \lambda_i^2. \quad (6)$$

This establishes the equivalence $\mathcal{L}(G) = \text{tr}(\mathbf{L}^2)$.

We now compute $\text{tr}(\mathbf{L}^2)$ directly from the elemental definition of $\mathbf{L}$. For a general weighted graph, let w_{ij} denote the weight of the edge between v_i and v_j (with $w_{ij} = 0$ if no edge exists). The entries of $\mathbf{L}$ are defined as:

$$L_{ij} = \begin{cases} \deg(v_i) & \text{if } i = j, \\ -w_{ij} & \text{if } i \neq j \text{ and } (v_i, v_j) \in E, \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

The trace of $\mathbf{L}^2$ is the sum of its diagonal entries, $\text{tr}(\mathbf{L}^2) = \sum_{i=1}^n (\mathbf{L}^2)_{ii}$. By the definition of matrix multiplication, a diagonal entry $(\mathbf{L}^2)_{ii}$ is:

$$(\mathbf{L}^2)_{ii} = \sum_{k=1}^n L_{ik}L_{ki} = \sum_{k=1}^n L_{ik}^2 \quad (\text{since } \mathbf{L} \text{ is symmetric}). \quad (8)$$

Therefore, the trace of $\mathbf{L}^2$ is the sum of squares of all entries in $\mathbf{L}$:

$$\text{tr}(\mathbf{L}^2) = \sum_{i=1}^n \sum_{k=1}^n L_{ik}^2. \quad (9)$$

We can decompose this sum into the sum of squares of the diagonal and off-diagonal entries:

1. **Diagonal entries:** The sum of squares of the diagonal entries is $\sum_{i=1}^n L_{ii}^2 = \sum_{v \in V} \deg(v)^2$.
2. **Off-diagonal entries:** For each edge $e = (v_i, v_j) \in E$ with weight $w_e = w_{ij}$, there are two non-zero off-diagonal entries in $\mathbf{L}$: $L_{ij} = -w_e$ and $L_{ji} = -w_e$. Thus, the sum of squares of off-diagonal entries is

$$\sum_{i \neq j} L_{ij}^2 = \sum_{(v_i, v_j) \in E} (L_{ij}^2 + L_{ji}^2) = \sum_{e \in E} ((-w_e)^2 + (-w_e)^2) = 2 \sum_{e \in E} w_e^2.$$

Combining these two parts gives a concrete expression for $\text{tr}(\mathbf{L}^2)$:

$$\text{tr}(\mathbf{L}^2) = \sum_{v \in V} \deg(v)^2 + 2 \sum_{e \in E} w_e^2. \quad (10)$$

By equating this with our earlier finding that $\mathcal{L}(G) = \text{tr}(\mathbf{L}^2)$, we arrive at the desired result:

$$\mathcal{L}(G) = \sum_{v \in V} \deg(v)^2 + 2 \sum_{e \in E} w_e^2. \quad (11) \quad \square$$

Remark. The decomposition confers substantial algorithmic benefits for reducing computational complexity, from an $O(n^3)$ spectral decomposition to an $O(n)$ degree aggregation. It also enables efficient energy adjustments for localized graph updates.